

Software Testing

Module 5

Module-End Assignment:

**“Integrated Tools for Software Development and
Project Management”**

1. Demonstrate a Sample API Testing using Postman API

- Perform API testing using Postman.
- Create a simple collection to test REST APIs (GET, POST, PUT, DELETE).
- Validate status codes, response body, headers, and response time.
- Use environment variables and assertions for dynamic validation.
- Generate and export test results in JSON or HTML format.

Postman is one of the most popular software testing tools which is used for **API** (Application Programming Interface) **testing**. It allows testers to send HTTP requests (GET, POST, PUT, DELETE) and verify API responses.

REST APIs are tested in Postman using “[API Playground - ReqRes](#)” website:

▪ Method 1: **GET**

Status Code: **200 OK**

Body in JSON:

```
{
  "page": 2,
  "per_page": 6,
  "total": 12,
  "total_pages": 2,
  "data": [
    {
      "id": 7,
      "email": "michael.lawson@reqres.in",
      "first_name": "Michael",
      "last_name": "Lawson",
      "avatar": "https://reqres.in/img/faces/7-image.jpg"
    },
    {
```

```
"id": 8,
"email": "lindsay.ferguson@reqres.in",
"first_name": "Lindsay",
"last_name": "Ferguson",
"avatar": "https://reqres.in/img/faces/8-image.jpg"
},
{
  "id": 9,
  "email": "tobias.funke@reqres.in",
  "first_name": "Tobias",
  "last_name": "Funke",
  "avatar": "https://reqres.in/img/faces/9-image.jpg"
},
{
  "id": 10,
  "email": "byron.fields@reqres.in",
  "first_name": "Byron",
  "last_name": "Fields",
  "avatar": "https://reqres.in/img/faces/10-image.jpg"
},
{
  "id": 11,
  "email": "george.edwards@reqres.in",
  "first_name": "George",
  "last_name": "Edwards",
  "avatar": "https://reqres.in/img/faces/11-image.jpg"
},
{
  "id": 12,
```

```
    "email": "rachel.howell@reqres.in",
    "first_name": "Rachel",
    "last_name": "Howell",
    "avatar": "https://reqres.in/img/faces/12-image.jpg"
  }
],
"support": {
  "url": "https://benhowdle.im/first-cto-
playbook?utm_source=reqres&utm_medium=json&utm_campaign=referral",
  "text": "Become a better CTO. A playbook of painful stories and practical advice from a
two-time startup CTO."
},
"_meta": {
  "powered_by": "ReqRes",
  "docs_url": "https://app.reqres.in/documentation",
  "upgrade_url": "https://app.reqres.in/upgrade",
  "example_url": "https://app.reqres.in/examples/notes-app",
  "variant": "v1_b",
  "message": "This is a read-only demo endpoint. Sign up to create your own collections
with full CRUD and auth.",
  "cta": {
    "label": "Get started",
    "url": "https://app.reqres.in/upgrade"
  },
  "context": "legacy_success"
}
}
```

Assertions used: **Response time** validation

Test Script:

```
pm.test("Response time is less than 200ms", function () {  
    pm.expect(pm.response.responseTime).to.be.below(200);  
});
```

Test Results:

FAILED: Response time is less than 200ms | AssertionError: expected 525 to be below 200

▪ Method 2: **POST**

Status Code: **200 OK** (201)

Body in JSON:

```
{  
  "id": 4,  
  "token": "QpwL5tke4Pnpja7X4",  
  "_meta": {  
    "powered_by": "ReqRes",  
    "docs_url": "https://app.reqres.in/documentation",  
    "upgrade_url": "https://app.reqres.in/upgrade",  
    "example_url": "https://app.reqres.in/examples/notes-app",  
    "variant": "v1_b",  
    "message": "This is a read-only demo endpoint. Sign up to create your own collections  
with full CRUD and auth.",  
    "cta": {  
      "label": "Get started",  
      "url": "https://app.reqres.in/upgrade"  
    },  
    "context": "legacy_success"  
  }  
}
```

```
}
```

Assertions used: **Response body** validation

Test Script:

```
pm.test("Body matches string", function () {  
    pm.expect(pm.response.text()).to.include("id");  
});
```

Test Results:

PASSED: Body matches string

▪ Method 3: **PUT**

Status Code: **200 OK**

Body in JSON:

```
{  
  "name": "morpheus",  
  "job": "zion resident",  
  "updatedAt": "2026-08-01T10:39:08.121Z",  
  "_meta": {  
    "powered_by": "ReqRes",  
    "docs_url": "https://app.reqres.in/documentation",  
    "upgrade_url": "https://app.reqres.in/upgrade",  
    "example_url": "https://app.reqres.in/examples/notes-app",  
    "variant": "v1_b",  
    "message": "This is a read-only demo endpoint. Sign up to create your own collections  
with full CRUD and auth.",  
    "cta": {  
      "label": "Get started",  
      "url": "https://app.reqres.in/upgrade"  
    }  
  },  
}
```

```
    "context": "legacy_success"
  }
}
```

Assertions used: **Response headers** validation

Test Script:

```
pm.test("Content-Type is present", function () {
    pm.response.to.have.header("Content-Type");
});
```

Test Results:

PASSED: Content-Type is present

▪ Method 4: **DELETE**

Status Code: **204 No Content**

Assertions used: **Status code** validation

Test Script:

```
pm.test("Status code is 204", function () {
    pm.response.to.have.status(204);
});
```

Test Results:

PASSED: Status code is 204

▪ **Environment Variables:**

Created an Environment Variable: **url**

Value: **https://reqres.in/api**

Instead of writing the complete URL, used like this: **{{url}}/users?page=2**

2. Building a Web Test Plan & HTML Report Generation (JMeter)

- Design and configure a Web Test Plan in Apache JMeter for a sample website.
- Include Thread Group, HTTP Request, and Listener components.
- Execute the test with defined users and ramp-up settings.
- Analyze test results (Response Time, Throughput, Error %).
- Export performance metrics and generate an HTML Summary Report.

JMeter is primarily used for performance testing, stress testing, and load testing for web applications, databases, and other services.

Test Plan Configuration

Thread Group

Number of Users: 5

Ramp-Up Period: 5 seconds

Loop Count: 1

HTTP Request Sampler

Server Name: www.saucedemo.com

Method: GET

Listener

Add:

- View Results in Table
- Summary Report
- Aggregate Report